YUJIE ZHU

Jersey City, NJ | yujie.zhu51@gmail.com | (516) 923 2404

EDUCATION

University at Buffalo, SUNY *(GPA: 3.96)* Buffalo, NY
Master of Science in Computer Science and Engineering June 2026
- Presidential Fellowship

New York University *(GPA: 3.90)* New York City, NY
Master of Science in Computer Engineering May 2024

Southern University of Science and Technology Shenzhen, China
Bachelor of Science in Applied Mathematics July 2022
- Outstanding Graduates of Mathematics Department

PROFESSIONAL EXPERIENCE

Aurora US Consulting LLC Jersey City, NJ
Co-Founder & Technical Consultant June 2026 –
- Co-founded an education and technology consulting company and managed technical operations, including website development and maintenance.
- Consulted with clients to assess their technical backgrounds, identify project needs, and develop appropriate software development and research plans.
- Mentored students in software development projects, providing guidance on programming fundamentals, project design, implementation, and debugging.
- Mentored undergraduate students in machine learning research, including literature reviews, research methodologies, and introductory research projects.

University at Buffalo, SUNY Buffalo, NY
Research Assistant Sept. 2024 – June 2026
- Conducted advanced research in machine learning and mathematical optimization, focusing on algorithmic convergence and complexity analysis.
- Investigated prompt-based continual learning methods to mitigate catastrophic forgetting.
- Investigated bilevel optimization algorithms, including warm-start strategies and single-loop approaches, to evaluate convergence behavior and computational efficiency.
- Investigated AdaGrad-Norm adaptive step sizes for FedAvg in federated learning.
- Analyzed theoretical lower bounds for bilevel optimization using zero-chain functions.

Head Teaching Assistant — Data Structures and Algorithms Sept. 2024 – June 2025
- Coordinated the teaching assistant team, developed grading criteria, and managed course operations for an undergraduate algorithms course.

PUBLICATIONS

- Zhu, Y. (2026). CLIP-Guided Label-Free Discriminative Region Scoring for Fine-Grained Classification. arXiv preprint, arXiv:2607.13437.
- Zhu, Y. (2021). Self-supervised Learning for Small Shot COVID-19 Classification. *2021 3rd International Conference on Information Technology and Computer Communications* (pp. 36–40). ACM ISBN 978-1-4503-8988-4/21/06.

TECHNICAL SKILLS

Areas of Expertise: Deep Learning, Computer Vision, Algorithm Analysis, Optimization
Programming: Python, Java, C++, MATLAB, R, LaTeX